Cloud height calculation using casted shadows

Team name: AstroX

Chosen theme: Life on Earth

Organisation name: Masaryk Grammar School Plzeň

Country: Czech Republic

1. Introduction

Our investigation regarded the ceiling height of clouds in Earth's atmosphere and the possibility of calculating it purely from satellite images. Our objective was to develop an algorithm, which could, with the help of the provided Coral chip, perform real-time calculation of the cloud height using clouds and their shadows.

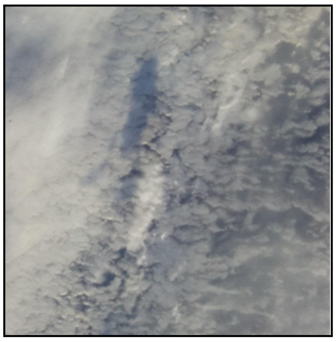

Fig. 1: Shadow of stratocumulus on stratus

Ceiling cloud height can be measured from the ground using laser ceilometers, alternatively by using satellites equipped with LiDAR e. g. NASA and CNES CALIPSO[1], or it can be also determined by calculating the dew point using air humidity and ground temperature. Nevertheless it is complicated, expensive, and sensor-heavy, albeit more accurate than our method.

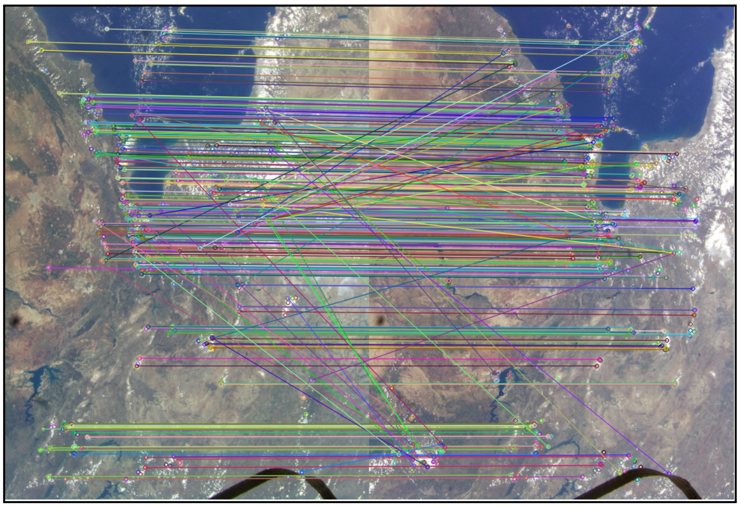

Fig. 2: Determination of camera rotation - similar features linked

We encountered many obstacles such as the ever changing location of north in the pictures, the uncertain direction which the camera was facing and even more technical problems like the need for parallel computing in order to ensure regular intervals between photographs.

We expected to find the calculated cloud height correlated to known cloud types and we also felt excited comparing the algorithmic data with the manually calculated ones.

[1] "Evaluation of cloud base height in the North American Regional ...." November 1, 2019, https://rmets.onlinelibrary.wiley.com/doi/full/10.1002/joc.6389. Access Date: June 21, 2023.

2. Method

To reach our objective the most important data source were images taken using the on-board HQ VIS camera, as a result of our azimuth calculation algorithm[2] we were also very dependent on embedded location data. We collected data from other sensors in case our main objective failed fatally which fortunately did not occur.

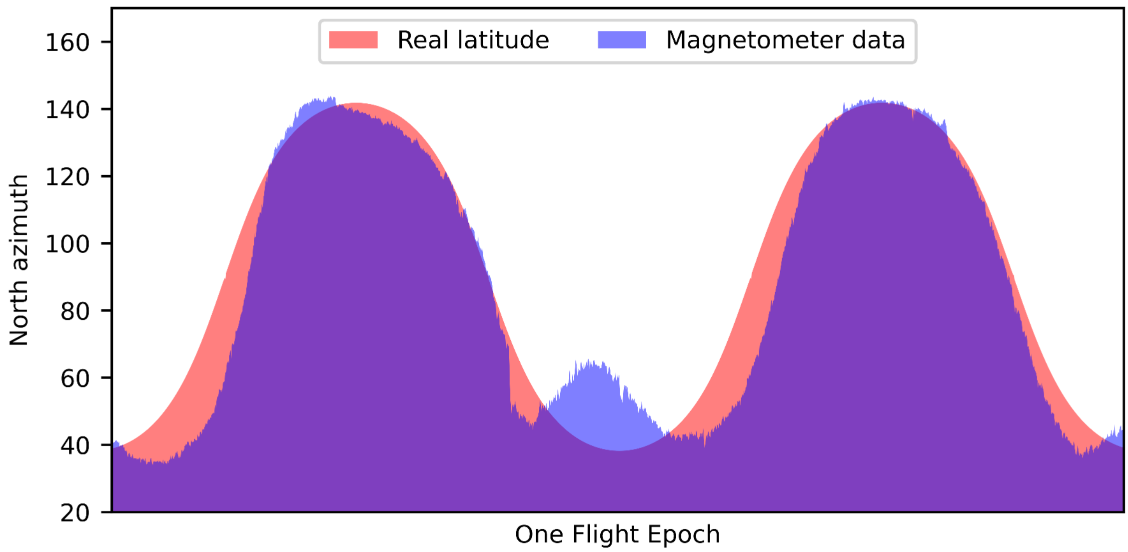

Fig. 3: Proof of magnetometer inaccuracy

Firstly for determining the rotation of the AstroPi camera, a specific algorithm operating on the basis of comparing similar pixel samples between two images (Fig. 2), was integrated into the code. This was then combined with the azimuth of north stemming from the sinusoidal orbit of the ISS rather than with the inexact data from the onboard compass, which can be seen on Fig. 3.

Images that were taken and saved were cropped and chopped into smaller sectors and then the individual clouds were detected using machine learning. For each cloud a special kind of pixel-readout fingerprint was created and on the forementioned curve the distance between shadow and cloud was established using four calculation methods: brightness and its first derivative with respect to distance in the RGB and red spectra. This distance then yielded cloud height using the altitude of the sun and trigonometrics.

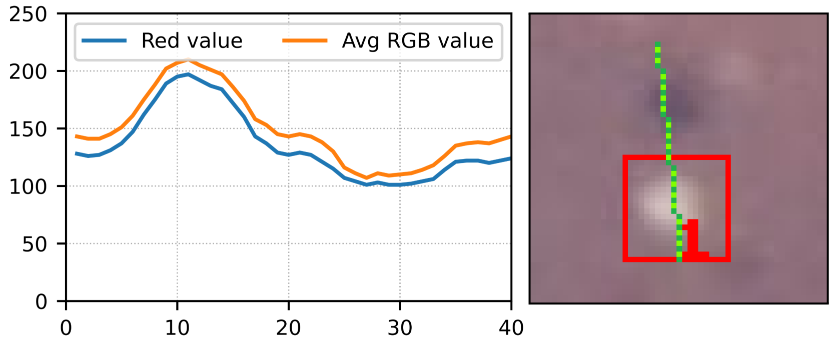

Fig. 4: Visualisation of pixel-readout fingerprint

3. Experiment results

The most important finding was that our program ran nominally. The calculation of north was accurate and cloud object detection turned out effective. Primarily the main thread captured 470 images, which occupied 2 GiB, whereof only a quarter were completely dark night shots.

[2] "GitHub: AstroX, matyasmatta · GitHub." https://github.com/matyasmatta/astrox/tree/report. Access Date: June 23,2023.

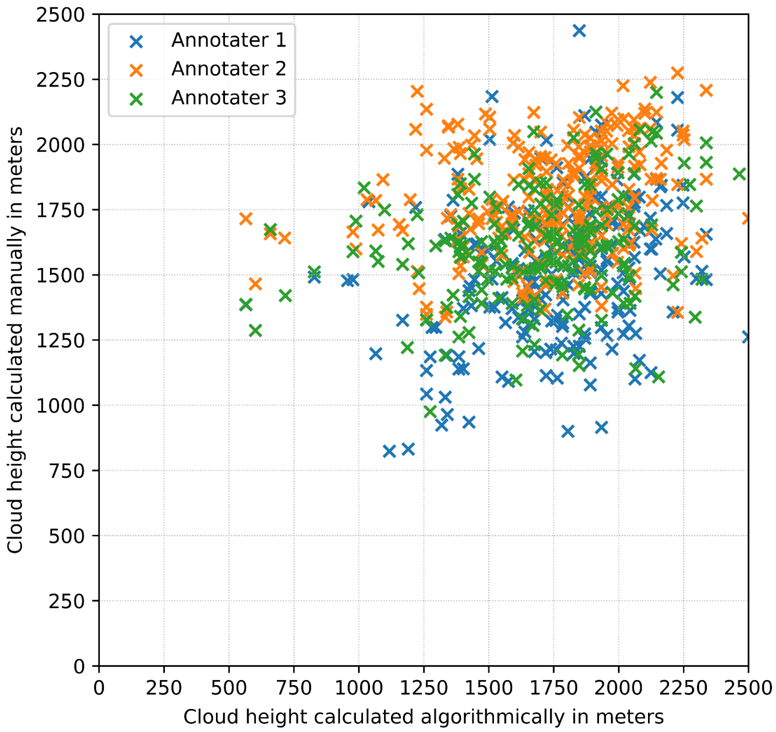

Fig. 5: Comparision of algorithmic and manual results

In further analysis there were major problems found with the shadow detection algorithm itself, as the photos we obtained were unpredictably distorted, completely impeding any accurate determination of the direction shadows should be detected in - all due to faulty positioning of the camera. We were forced to use only 8% of every image.

Distortion difficulties were partially overcome by quadrilateral correction, and the disregard for image sectors far from ISS's current location. Although we conceptually formulated a very accurate correction mechanism compensating even for the differences of cloud placement relative to the station, we did not manage to accomplish it within the tight schedule.

The data received from the ISS were more thoroughly investigated back on Earth, where we also trained a new ML YOLOv8 based model, which greatly improved the accuracy and number of detections to 300 clouds per sector, thirty times more than the TFLite model ran on the ISS. In order to validate our data we developed an application specifically tailored to manually determine cloud height in our images, this data were later compared to the algorithmically calculated predictions (Fig. 5). If the shadow detection algorithm was as accurate as we were, data would concentrate in a cluster near the line of equality.

The data analysis itself proved to be very challenging as clouds on which our detection algorithms were successful, usually cumuli, as they cast the finest shadow, were sporadically interspersed in our dataset.

4. Learnings

When our experiment was approved, we struggled with structuring our work. At first, we wanted to work remotely, however, this proved impossible, ergo we contacted the University of West Bohemia[3], a local university willing to provide us with a laboratory for our work. This had an enormous impact on our collaboration as we could now discuss and solve problems together. Over time each of us developed an expertise for a specific project component, thus naturally splitting up work.

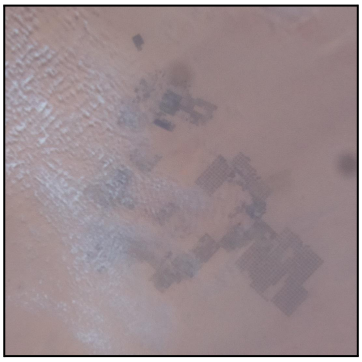

Fig. 6: Circular fields in Saudi Arabia

[3] "About us - University of West Bohemia." https://www.zcu.cz/en/University/About-us/index.html. Access Date: June 23, 2023.

Fig. 7: Flight path from daylight images

The biggest challenges came in the beginning when we encountered numerous unsolved problems; these required complex insight that we had to obtain, for example, the camera orientation. With that came the gradual improvement of our coding abilities.

Aside from that, we also managed to further improve our presentation skills, as we held a public lecture presenting the project at the University of West Bohemia. The next goal is to publish a scientific article depicting our experiment.

For the upcoming year we might select a bit more trivial topic, considering that we have spent approximately 160 hours per person on this project. Moreover we could take more images during the day since we did use a third of the capacity.

5. Conclusion

Although our cloud detection system might not be the most accurate and effective solution, the methods we used to achieve it opened up a whole new realm of possible uses. We were informed that our height calculation algorithm might be applicable not only in meteorology but also in archaeology, and that our correction algorithms might prove beneficial in geomatics. The results and mainly the successful run of our code made us very content, and although it was not perfect, it shall serve as a proof of concept.

Fig. 8: Our lecture was reported on by local news

We all experienced a great time collaborating, we gained an unfathomable amount of knowledge which we will indubitably apply during our future endeavours. Despite many hours of frustration over malfunctioning code and unsolvable problems, we have managed to create something we are proud of. We could have never foreseen how far this competition would bring us, for instance at the university's "Take Your Child to Work Day".

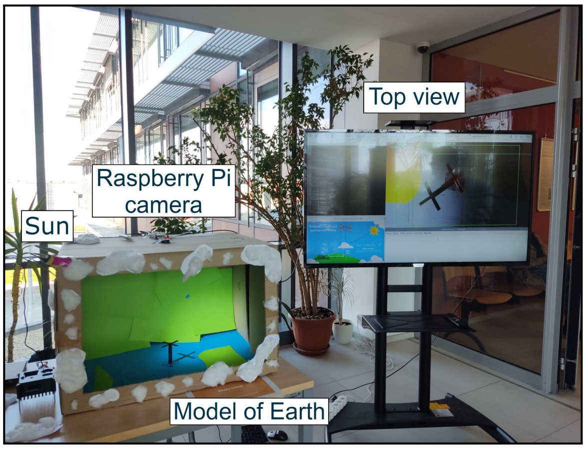

Fig. 9: Demonstration at Bring Your Child to Work Day

Our gratitude shall go to the University of West Bohemia (mainly the Faculty of Applied Sciences) for providing us with their facilities, our school[4] which allowed us to pursue our ambitions and Planetum Prague[5] for introducing us to AstroPi via a hackathon.

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[4] "General information - Masarykovo gymnázium Plzeň." https://www.mgplzen.cz/our-activities/general-information. Access Date: June 23, 2023.

[5] "Štefánik Observatory - Planetum." https://www.planetum.cz/stefanik_observatory/. Access Date: June 23, 2023.